		nT | $\frac{1}{2}g(nT)^2 = n^2\left(\frac{1}{2}gT^2\right)$
7	**A**	$P = Fv$ $0.4 \times 800 = F \times 13$ Force applied by car F = 24.6 N Constant velocity, for F = resistive force = 25 N
8	**D**	Centripetal force $F_c = mr\omega^2$ is provided by the friction between the object and the turntable.